The Renormalization Splitting

QED and QCD Side by Side

PHYS 7502 — Supplementary Notes

Goal

Show explicitly how one loop integral — which is divergent — gets split into an **infinite but unobservable** piece that redefines a coupling constant, plus a **finite and predictive** piece that gives the running of that coupling. Do this for both QED (photon self-energy) and QCD (gluon self-energy).

Part I: QED — Vacuum Polarization

Setup: the one-loop photon propagator

At tree level the photon propagator is

$$D_0^{\mu\nu}(q) \;=\; \frac{-ig^{\mu\nu}}{q^2}.$$

At one loop, a virtual e^+e^- pair dresses the propagator. The corrected propagator becomes

$$D^{\mu\nu}(q) \;=\; \frac{-ig^{\mu\nu}}{q^2}\,\frac{1}{1-e_0^2\,\Pi(q^2)},$$

where $\Pi(q^2)$ is the **photon self-energy** (the loop integral, stripped of the bare propagators). The effective coupling strength at the photon line is

$$\boxed{e^2(q^2) \;=\; \frac{e_0^2}{1-e_0^2\,\Pi(q^2)}}$$

Problem: $\Pi(q^2)$ diverges. It contains a term that blows up when the loop momentum $k \to \infty$.

Step 1: Write the loop integral

The one-loop self-energy is (schematically, after Feynman parametrization and Wick rotation):

$$\Pi(q^2) \;=\; -\frac{1}{2\pi^2}\int_0^1 dx\; x(1-x)\;\ln\!\left[\frac{\Lambda^2}{m^2 - x(1-x)\,q^2}\right]$$

where Λ is an ultraviolet cutoff regulating the divergence, and m is the electron mass.

The $\ln\Lambda^2$ term is the divergence: it grows without bound as $\Lambda \to \infty$.

Step 2: The splitting (the key move)

Now do the most innocent algebraic step imaginable — evaluate Π at $q^2 = 0$ and subtract:

The splitting identity

$$\Pi(q^2) \;=\; \underbrace{\Pi(0)}_{\substack{\text{divergent}\\ \text{(no } q^2 \text{ dependence)}}} \;+\; \underbrace{\big[\Pi(q^2)-\Pi(0)\big]}_{\substack{\text{finite}\\ \text{(all the } q^2 \text{ dependence)}}}$$

This is just $A = B + (A - B)$. But look at each piece:

The divergent piece $\Pi(0)$:

$$\Pi(0) \;=\; -\frac{1}{2\pi^2}\int_0^1 dx\; x(1-x)\;\ln\!\Big[\frac{\Lambda^2}{m^2}\Big] \;=\; -\frac{1}{12\pi^2}\;\ln\!\Big[\frac{\Lambda^2}{m^2}\Big]$$

This contains $\ln\Lambda^2$ and diverges as $\Lambda\to\infty$. But it is a **pure number** — it has *zero* dependence on q^2.

The finite piece $\Pi(q^2)-\Pi(0)$:

$$\Pi(q^2)-\Pi(0) \;=\; -\frac{1}{2\pi^2}\int_0^1 dx\; x(1-x)\;\ln\!\Big[\frac{m^2}{m^2-x(1-x)\,q^2}\Big]$$

The $\ln\Lambda^2$ terms in $\Pi(q^2)$ and $\Pi(0)$ are **identical** (same integrand at high k), so they **cancel** in the subtraction. What remains depends on q^2 and is finite.

Step 3: Absorb the divergent piece into the charge

Substitute the splitting into the effective coupling:

$$e^2(q^2) \;=\; \frac{e_0^2}{1-e_0^2\,\Pi(0)-e_0^2[\Pi(q^2)-\Pi(0)]}$$

Now **define the physical (renormalized) charge** as the coupling measured at $q^2 = 0$:

$$\boxed{e_R^2 \;\equiv\; \frac{e_0^2}{1-e_0^2\,\Pi(0)}}$$

Both e_0 and $\Pi(0)$ are individually infinite, but e_R is what your ammeter reads — it's finite by definition (it's a measured quantity: $e_R^2/4\pi = \alpha \approx 1/137$).

Step 4: Re-express everything in terms of e_R

After absorbing $\Pi(0)$ into e_R, the effective coupling at scale q^2 becomes:

$$e^2(q^2) \;=\; \frac{e_R^2}{1-e_R^2\big[\Pi(q^2)-\Pi(0)\big]}$$

Every quantity on the right is **finite**:

- e_R^2 is measured.
- $\Pi(q^2)-\Pi(0)$ is a finite integral.
- Λ is gone — it was absorbed into e_R.

Step 5: Evaluate the finite piece

In the high-energy limit $|q^2| \gg m^2$, the integral gives:

$$\Pi(q^2) - \Pi(0) \;\approx\; \frac{1}{12\pi^2}\,\ln\!\left[\frac{|q^2|}{m^2}\right]$$

Substituting into the running coupling and writing $\alpha = e_R^2/4\pi$:

QED running coupling

$$\boxed{\alpha(q^2) \;=\; \frac{\alpha(\mu^2)}{1 \;-\; \dfrac{\alpha(\mu^2)}{3\pi}\,\ln\!\left(\dfrac{|q^2|}{\mu^2}\right)}}$$

- The **minus sign** in the denominator means α **increases** with $|q^2|$. (Screening: you penetrate the virtual-pair cloud.)
- At low energies: $\alpha \approx 1/137$.
- At $q^2 = M_Z^2$: $\alpha \approx 1/128$.

Physical meaning of the QED splitting

- $\Pi(0)$ = the infinite dressing that turns the "bare" charge into the measured charge. You can never see e_0 in isolation — every measurement already includes this dressing. So absorbing $\Pi(0)$ into e_R isn't a trick; it's the acknowledgment that e_0 was never the physical parameter.
- $\Pi(q^2) - \Pi(0)$ = the *difference* in dressing between scale q^2 and scale 0. This is observable: it tells you how the effective charge changes as you vary the momentum transfer. That's the running coupling.

Part II: QCD — Gluon Self-Energy

Setup: why QCD is different

The splitting procedure is **identical in structure** to QED. What changes is the *content* of the loop.
In QED, only one type of diagram contributes to the photon self-energy:

virtual e^+e^- loop (fermion loop)

In QCD, **three** types of diagram contribute to the gluon self-energy:

(a) quark loop (same structure as QED)
(b) gluon loop (gluon self-interaction — **new!**)
(c) ghost loop (technical fix for gauge redundancy)

Diagrams (b) and (c) exist because gluons carry color charge. Photons don't carry electric charge, so QED has no analog.

Step 1: Write the loop integral

The gluon self-energy $\Pi_{\rm QCD}(q^2)$ receives contributions from all three diagram types. After regularization, each contributes a term proportional to $\ln\left(\Lambda^2/\mu^2\right)$ (divergent) plus a q^2-dependent finite piece.

Schematically:

$$\Pi_{\rm QCD}(q^2) \;=\; \underbrace{-\frac{n_f}{6\pi^2}\ln\frac{\Lambda^2}{\mu^2}}_{\text{quark loops (screening)}} \;+\; \underbrace{+\frac{11C_A}{48\pi^2}\ln\frac{\Lambda^2}{\mu^2}}_{\text{gluon + ghost loops (anti-screening)}} \;+\; [\text{finite } q^2\text{-dependent terms}]$$

where n_f is the number of active quark flavors and $C_A = N_c = 3$ for SU(3).

Step 2: The same splitting identity

Exactly as in QED:

$$\Pi_{\rm QCD}(q^2) \;=\; \Pi_{\rm QCD}(0) \;+\; \left[\Pi_{\rm QCD}(q^2) - \Pi_{\rm QCD}(0)\right]$$

- $\Pi_{\rm QCD}(0)$: divergent, q^2-independent $\longrightarrow$ absorbed into g_s.
- $\Pi_{\rm QCD}(q^2) - \Pi_{\rm QCD}(0)$: finite, all the q^2-dependence $\longrightarrow$ gives $\alpha_s(q^2)$.

Step 3: Absorb the divergent piece into g_s

Define the renormalized strong coupling $g_{s,R}$ at reference scale μ^2:

$$g_{s,R}^2 \;\equiv\; \frac{g_{s,0}^2}{1 - g_{s,0}^2\,\Pi_{\rm QCD}(0)}$$

Same logic: $g_{s,0}$ and $\Pi_{\rm QCD}(0)$ are each infinite; $g_{s,R}$ is what a measurement at scale μ^2 returns.

Step 4: Evaluate the finite piece

The crucial difference from QED: the **quark loops** and the **gluon loops** contribute with **opposite signs** to the finite remainder. The combined coefficient is:

$$\Pi_{\text{QCD}}(q^2) - \Pi_{\text{QCD}}(\mu^2) \;\approx\; \frac{1}{12\pi^2}\Big(\underbrace{11C_A}_{\substack{\text{gluon loops}\\ \text{anti-screening}\\ =+33}} \;-\; \underbrace{2n_f}_{\substack{\text{quark loops}\\ \text{screening}}} \Big)\, \ln\Big[\frac{|q^2|}{\mu^2}\Big]$$

The magic number is $(33 - 2n_f)$. For 6 quark flavors: $(33 - 12) = 21 > 0$. In fact, for *any* $n_f \leq 16$, this coefficient is positive.

Step 5: The QCD running coupling

QCD running coupling

$$\boxed{\alpha_s(q^2) \;=\; \frac{\alpha_s(\mu^2)}{1 \;+\; \dfrac{\alpha_s(\mu^2)}{12\pi}\,(33 - 2n_f)\,\ln\Big(\dfrac{|q^2|}{\mu^2}\Big)}}$$

- The **plus sign** in the denominator (opposite to QED!) means α_s **decreases** with $|q^2|$.
- High energy (large $|q^2|$): $\alpha_s \to 0$. This is **asymptotic freedom**.
- Low energy (small $|q^2|$): α_s grows toward ~ 1. Perturbation theory breaks down $\to$ **confinement**.

Part III: Comparison — Why the Sign Flips

Side-by-side summary

QED	**QCD**
Loop content: fermion (e^+e^-) only	Loop content: quarks + gluons + ghosts
Divergent piece $\Pi(0)$: absorbed into e_R	Divergent piece $\Pi(0)$: absorbed into $g_{s,R}$
Finite piece: $+\frac{1}{12\pi^2}\ln\frac{\lvert q^2\rvert}{\mu^2}$	Finite piece: $+\frac{33-2n_f}{12\pi^2}\ln\frac{\lvert q^2\rvert}{\mu^2}$
Denominator: $1-\frac{\alpha}{3\pi}\ln\frac{\lvert q^2\rvert}{\mu^2}$	Denominator: $1+\frac{\alpha_s(33-2n_f)}{12\pi}\ln\frac{\lvert q^2\rvert}{\mu^2}$
Minus sign $\to$ α **grows**	Plus sign $\to$ α_s **shrinks**
Screening	**Anti-screening wins**
Coupling at M_Z: $\alpha \approx 1/128$	Coupling at M_Z: $\alpha_s \approx 0.118$
Coupling at $q^2 = 0$: $\alpha = 1/137$	Coupling at $q^2 \to 0$: $\alpha_s \to \infty$ (confinement)

The one-paragraph summary

In both QED and QCD, you compute a self-energy loop that diverges. You split it into a q^2-independent divergent constant (which redefines the coupling you measure) and a q^2-dependent finite remainder (which is the running). The splitting is the same algebraic identity in both theories: $\Pi(q^2) = \Pi(0) + [\Pi(q^2) - \Pi(0)]$. The *physics* differs because QCD's non-abelian gluon self-coupling introduces loop diagrams with no QED analog, and those diagrams flip the sign of the finite remainder. That sign flip is the entire difference between screening (QED) and anti-screening (QCD), and therefore between a theory where coupling grows at short distances and one where coupling vanishes at short distances.